

# Probability basics

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## Defining probabilities

**Definition 0.1** (Sample space). The **sample space** of a random experiment, denoted  $\Omega$ , is the set of all of its possible outcomes.

**Example 0.1** (Sample space of a die roll). Rolling a six-sided die once has sample space  $\Omega = \{1, 2, 3, 4, 5, 6\}$ .

**Definition 0.2** (Event). An **event** is a subset of the **sample space**  $\Omega$ : a set of outcomes. An event  $A$  **occurs** when the experiment's outcome is in  $A$ .

When  $\Omega$  is uncountable, such as an interval of real numbers, only the subsets in a designated collection  $\mathcal{S}$  (a  **$\sigma$ -algebra**: a collection that contains  $\Omega$  and is closed under complements and countable unions) count as events; every set that arises in these notes is in  $\mathcal{S}$ .

**Example 0.2** (Rolling an even number). In Example 0.1, the event “the roll is even” is  $A = \{2, 4, 6\}$ .

**Definition 0.3** (Complement of an event). The **complement** of an **event**  $A$ , denoted  $\neg A$ , is the event that  $A$  does not occur: the set of outcomes in the sample space  $\Omega$  that are not in  $A$ .

$$\neg A \stackrel{\text{def}}{=} \Omega \setminus A$$

Other sources write  $A^c$  or  $\bar{A}$ .

**Example 0.3** (Rolling an odd number). In Example 0.2, the complement of the event “the roll is even”,  $A = \{2, 4, 6\}$ , is the event “the roll is odd”,  $\neg A = \{1, 3, 5\}$ .

**Definition 0.4** (Mutually exclusive events). Finitely or countably many events  $A_1, A_2, \dots$  are **mutually exclusive** (also called **disjoint**, **pairwise disjoint**, or **mutually disjoint**) when no two of them share an outcome:

$$A_i \cap A_j = \emptyset \quad \text{for all } i \neq j$$

**Example 0.4** (Low and high rolls). In Example 0.1, the events “the roll is at most 2”,  $\{1, 2\}$ , and “the roll is at least 5”,  $\{5, 6\}$ , are mutually exclusive: no roll is in both. The events “the roll is even”,  $\{2, 4, 6\}$ , and “the roll is at most 2”,  $\{1, 2\}$ , are not mutually exclusive: a roll of 2 is in both.

**Definition 0.5** (Partition of an event). A **partition** of an event  $A$  is a finite or countably infinite collection of **mutually exclusive** events  $A_1, A_2, \dots$  whose union is  $A$ :

$$\bigcup_i A_i = A$$

A partition of the **sample space** is the case  $A = \Omega$ .

Some sources also require each piece  $A_i$  to contain at least one outcome.

**Example 0.5** (Partitioning the die roll). In Example 0.1, the events “the roll is even”,  $\{2, 4, 6\}$ , and “the roll is odd”,  $\{1, 3, 5\}$ , are mutually exclusive, and their union is  $\Omega$ , so they partition the sample space. So do the six single-outcome events  $\{1\}, \{2\}, \dots, \{6\}$ . The events  $\{1, 2\}$  and  $\{5, 6\}$  from Example 0.4 do not partition  $\Omega$ : their union leaves out 3 and 4.

**Definition 0.6** (Finite additivity). A function  $\mu$  that assigns a real number  $\mu(A)$  to each event  $A$  is **finitely additive** if, for every finite collection of **mutually exclusive** events  $A_1, \dots, A_n$ , the value of their union is the sum of their values:

$$\mu(A_1 \cup \dots \cup A_n) = \sum_{i=1}^n \mu(A_i)$$

**Example 0.6** (Counting outcomes is finitely additive). For the die roll in Example 0.1, let  $\mu(A) \stackrel{\text{def}}{=} |A|$ , the number of outcomes in  $A$ . The events  $\{1, 2\}$  and  $\{5, 6\}$  are mutually

exclusive (Example 0.4), and:

$$\begin{aligned}
 \mu(\{1, 2\} \cup \{5, 6\}) &= \mu(\{1, 2, 5, 6\}) && \text{(take the union)} \\
 &= 4 && \text{(count the outcomes)} \\
 &= 2 + 2 && \text{(write 4 as a sum)} \\
 &= \mu(\{1, 2\}) + \mu(\{5, 6\}) && \text{(count each event's outcomes)}
 \end{aligned}$$

The same holds for any mutually exclusive events, because the sizes of disjoint sets add, so  $\mu$  is finitely additive.

**Definition 0.7** (Countable additivity). A function  $\mu$  that assigns a real number  $\mu(A)$  to each event  $A$  is **countably additive** (also called  **$\sigma$ -additive**) if, for every sequence of mutually exclusive events  $A_1, A_2, \dots$ , the value of their union is the sum of their values:

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

Countable additivity implies finite additivity. Applying countable additivity to  $\emptyset, \emptyset, \dots$ , which are mutually exclusive with union  $\emptyset$ , gives  $\mu(\emptyset) = \sum_{i=1}^{\infty} \mu(\emptyset)$ , a sum of identical terms that is finite only if  $\mu(\emptyset) = 0$ . Then, for mutually exclusive  $A_1, \dots, A_n$ , set  $A_{n+1} = A_{n+2} = \dots = \emptyset$  and apply countable additivity to the whole sequence. The converse fails: there exist set functions that satisfy finite additivity but fail countable additivity (see [Wikipedia: Sigma-additive set function — An additive function which is not  \$\sigma\$ -additive](#)).

**Example 0.7** (Counting outcomes is countably additive). For the counting function  $\mu(A) \stackrel{\text{def}}{=} |A|$  of Example 0.6, take any sequence of mutually exclusive events  $A_1, A_2, \dots$ . No two of them share an outcome, and  $\Omega$  has only six outcomes, so at most six of the  $A_i$  contain any outcomes; the rest are  $\emptyset$ , with  $\mu(\emptyset) = 0$ . The infinite sum therefore has at most six nonzero terms, and finite additivity (Example 0.6) shows that those terms add up to  $\mu$  of the union. So  $\mu$  is countably additive.

**Definition 0.8** (Probability measure). A **probability measure** on a sample space  $\Omega$ , often denoted  $\Pr()$  or  $P()$ , is a function that assigns a number  $\Pr(A)$  to each event  $A$  and satisfies:

1. For any event  $A$ ,  $\Pr(A) \geq 0$ .
2. The probability of the whole sample space is 1:

$$\Pr(\Omega) = 1$$

3. Pr is **countably additive**: for any **mutually exclusive** events  $A_1, A_2, \dots$ ,

$$\Pr\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \Pr(A_i)$$

Property 3 implies that Pr is also **finitely additive**. Requiring countable additivity, not just finite additivity, enables results such as the continuity of probability (if  $A_1 \supseteq A_2 \supseteq \dots$  with  $\bigcap_i A_i = \emptyset$ , then  $\Pr(A_i) \rightarrow 0$ ), and it is what makes sums over countably infinite **partitions** valid.

**Example 0.8** (Probability measure for a fair die). For the die roll in Example 0.1, define  $\Pr(A) \stackrel{\text{def}}{=} |A|/6$ , where  $|A|$  is the number of outcomes in  $A$ . This function is non-negative, gives  $\Pr(\Omega) = 6/6 = 1$ , and is **countably additive**, because it is 1/6 times the counting function of Example 0.7. The event “the roll is even” from Example 0.2 has probability  $\Pr(\{2, 4, 6\}) = 3/6 = 1/2$ .

**Theorem 0.1** (Probability of a subset’s intersection). *If  $A$  and  $B$  are events and  $A \subseteq B$ , then  $\Pr(A \cap B) = \Pr(A)$ .*

**i** Proof

*Proof.* Since  $A \subseteq B$ , every outcome in  $A$  is also in  $B$ , so  $A \cap B = A$ , and therefore  $\Pr(A \cap B) = \Pr(A)$ . □

**Theorem 0.2** (An event and its complement sum to 1). *For any event  $A$  and its **complement**  $\neg A$ :*

$$\Pr(A) + \Pr(\neg A) = 1$$

**i** Proof

*Proof.* The events  $A$  and  $\neg A$  are disjoint, and their union is  $\Omega$ .

$$\begin{aligned} \Pr(A) + \Pr(\neg A) &= \Pr(A \cup \neg A) && \text{(additivity of probability for disjoint events)} \\ &= \Pr(\Omega) && (A \cup \neg A = \Omega) \\ &= 1 && \text{(probability of the sample space is 1)} \end{aligned}$$

□

**Corollary 0.1** (Complement rule). *For any event  $A$ :*

$$\Pr(\neg A) = 1 - \Pr(A)$$

**i** Proof

*Proof.* Subtract  $\Pr(A)$  from both sides of Theorem 0.2. □

**Corollary 0.2** (Complement rule in probability ( $\pi$ ) notation). *If the probability of an event  $A$  is  $\Pr(A) = \pi$ , then the probability that  $A$  does not occur is:*

$$\Pr(\neg A) = 1 - \pi$$

**i** Proof

*Proof.*

$$\begin{aligned}\Pr(\neg A) &= 1 - \Pr(A) && \text{(complement rule)} \\ &= 1 - \pi && \text{(substitute } \Pr(A) = \pi \text{)}\end{aligned}$$

□

**Example 0.9** (Probability of not rolling a six). For a fair die, the event “the roll is a six” has probability  $\pi = 1/6$ , so by Corollary 0.2 the probability of not rolling a six is  $1 - 1/6 = 5/6$ .

## Conditional probability

**Definition 0.9** (Conditional probability). For two events  $A$  and  $B$  with  $\Pr(B) > 0$ , the **conditional probability** of  $A$  given  $B$ , denoted  $\Pr(A \mid B)$ , is:

$$\Pr(A \mid B) \stackrel{\text{def}}{=} \frac{\Pr(A \cap B)}{\Pr(B)}$$

**Example 0.10** (Rolling a six, given an even roll). For a fair die (Example 0.8), let  $A = \{6\}$  and  $B = \{2, 4, 6\}$ . Then  $A \cap B = \{6\}$ , so:

$$\begin{aligned}
\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\
&= \frac{1/6}{3/6} && \text{(substitute the probabilities)} \\
&= \frac{1}{3} && \text{(simplify)}
\end{aligned}$$

**Theorem 0.3** (Law of conditional probability). *For any two events  $A$  and  $B$  with  $\Pr(B) > 0$ :*

$$\Pr(A \cap B) = \Pr(A \mid B) \cdot \Pr(B)$$

### **i** Proof

*Proof.* Rearranging Definition 0.9:

$$\begin{aligned}
\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\
\Pr(A \cap B) &= \Pr(A \mid B) \cdot \Pr(B) && \text{(multiply both sides by } \Pr(B))
\end{aligned}$$

□

**Example 0.11** (Applying the law of conditional probability). Suppose 30% of adults exercise regularly ( $\Pr(E) = 0.30$ ), and among adults who exercise regularly, 60% have low blood pressure ( $\Pr(L \mid E) = 0.60$ ).

Then, by Theorem 0.3, the probability that a randomly selected adult both exercises regularly and has low blood pressure is:

$$\begin{aligned}
\Pr(L \cap E) &= \Pr(L \mid E) \cdot \Pr(E) && \text{(law of conditional probability)} \\
&= 0.60 \cdot 0.30 && \text{(substitute the given values)} \\
&= 0.18 && \text{(multiply)}
\end{aligned}$$

**Theorem 0.4** (Law of total probability). *If  $B_1, B_2, \dots$  is a *partition* of the sample space, with  $\Pr(B_i) > 0$  for every  $i$ , then for any event  $A$ :*

$$\Pr(A) = \sum_i \Pr(A \mid B_i) \cdot \Pr(B_i)$$

**i** Proof

*Proof.* Since  $B_1, B_2, \dots$  partition the sample space, the events  $A \cap B_1, A \cap B_2, \dots$  are mutually exclusive and their union is  $A$ . By countable additivity (Definition 0.7), and then by Theorem 0.3:

$$\begin{aligned}\Pr(A) &= \sum_i \Pr(A \cap B_i) && \text{(countable additivity for partition of } A\text{)} \\ &= \sum_i \Pr(A \mid B_i) \cdot \Pr(B_i) && \text{(law of conditional probability; } \Pr(B_i) > 0\text{)}\end{aligned}$$

□

**Theorem 0.5** (Bayes' theorem). *For any two events  $A$  and  $B$  with  $\Pr(A) > 0$  and  $\Pr(B) > 0$ :*

$$\Pr(A \mid B) = \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)}$$

**i** Proof

*Proof.* By Definition 0.9 and Theorem 0.3:

$$\begin{aligned}\Pr(A \mid B) &= \frac{\Pr(A \cap B)}{\Pr(B)} && \text{(definition of conditional probability)} \\ &= \frac{\Pr(B \cap A)}{\Pr(B)} && \text{(intersection is commutative: } A \cap B = B \cap A\text{)} \\ &= \frac{\Pr(B \mid A) \cdot \Pr(A)}{\Pr(B)} && \text{(law of conditional probability applied to } \Pr(B \cap A)\text{)}\end{aligned}$$

□

**Example 0.12** (Positive predictive value of a medical test). Suppose a disease test has 99% sensitivity and 99% specificity, and the prevalence of the disease in the population is 7%.

Let  $D$  be the event “person has the disease” and  $+$  be the event “test is positive”. Then:

- $\Pr(+ \mid D) = 0.99$  (sensitivity)
- $\Pr(\neg+ \mid \neg D) = 0.99$  (specificity), so the false positive rate is  $\Pr(+ \mid \neg D) = 1 - 0.99 = 0.01$
- $\Pr(D) = 0.07$  (prevalence)

By Theorem 0.5, with the denominator expanded by the [law of total probability](#) over the

partition  $\{D, \neg D\}$ :

$$\begin{aligned}\Pr(D \mid +) &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+)} && \text{(Bayes' theorem)} \\ &= \frac{\Pr(+ \mid D) \cdot \Pr(D)}{\Pr(+ \mid D) \cdot \Pr(D) + \Pr(+ \mid \neg D) \cdot \Pr(\neg D)} && \text{(law of total probability)} \\ &= \frac{0.99 \cdot 0.07}{0.99 \cdot 0.07 + 0.01 \cdot 0.93} && \text{(substitute the given values)} \\ &= \frac{0.0693}{0.0693 + 0.0093} && \text{(multiply each term in the numerator and denominator by 0.01)} \\ &= \frac{0.0693}{0.0786} && \text{(add the denominator's two terms)} \\ &\approx 0.88 && \text{(divide)}\end{aligned}$$

Even with a highly accurate test (99% sensitive and 99% specific), only about 88% of people who test positive actually have the disease, because the disease prevalence is relatively low (7%).